

AITUTAKI, Cook Islands

WMO: 918310

Lat: 18.833S

Lon: 159.767W

Elev: 5

StdP: 101.26

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.1	20.8	12.4	9.0	21.6	13.4	9.6	21.9	11.4	23.0	10.5	23.3	3.8	140	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.8	30.3	26.3	29.9	26.1	29.4	25.8	27.2	29.4	26.7	28.9	26.4	28.6	4.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.0	28.8	26.1	21.5	28.4	25.7	21.0	28.1	85.7	29.5	84.0	29.2	82.4	28.8	34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.3	8.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	26.4	27.1	27.1	26.5	25.4	24.6	24.2	23.6	24.1	24.8	25.9	26.2
	DBStd	1.65	1.17	1.19	1.02	1.11	1.16	1.23	1.33	1.38	1.19	1.10	1.08	1.08
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5652	509	479	530	495	477	437	440	423	424	458	477	503
	CDD18.3	2610	251	246	272	245	219	187	182	164	174	199	227	244
	CDH23.3	18485	2168	2401	2627	2146	1439	924	809	563	703	1039	1674	1990
(14) Wind	WSAvg	4.0	3.6	3.9	3.0	3.8	4.0	4.0	4.8	4.1	4.2	4.5	4.2	3.9
	PrecAvg	1964	258	235	213	228	185	89	78	93	84	117	153	225
	PrecMax	2767	508	499	490	604	495	268	214	288	433	333	409	550
	PrecMin	1069	31	32	27	54	16	12	11	5	8	19	18	4
	PrecStd	396	115	111	112	132	121	71	56	74	86	72	98	127
	DB	30.4	30.9	30.9	30.3	29.1	28.5	28.2	28.2	28.3	28.8	29.9	30.2	30.2
	MCWB	26.5	26.4	27.0	26.3	25.7	25.4	25.2	24.9	24.5	24.4	24.8	26.4	26.4
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.7	30.2	30.2	29.6	28.4	27.6	27.4	27.2	27.6	28.0	29.2	29.3	29.3
	2% MCWB	26.2	26.3	26.5	25.9	25.2	24.5	24.2	24.1	23.8	24.0	24.9	25.8	25.8
	5% DB	29.1	29.7	29.7	29.1	27.9	27.1	26.9	26.5	27.0	27.3	28.6	28.8	28.8
	5% MCWB	25.9	26.1	26.2	25.6	24.8	24.0	23.7	23.5	23.4	23.5	24.7	25.4	25.4
	10% DB	28.6	29.2	29.1	28.5	27.2	26.6	26.2	25.8	26.2	26.7	27.9	28.2	28.2
	10% MCWB	25.5	25.7	25.9	25.2	24.3	23.7	23.4	22.8	22.8	23.1	24.4	25.0	25.0
	WB	27.3	27.2	27.7	27.4	26.5	26.1	25.7	25.6	25.5	25.7	26.6	27.4	27.4
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.6	29.7	30.1	29.6	28.3	27.8	27.5	27.3	27.3	27.6	28.5	29.2	29.2
	2% WB	26.7	26.8	27.1	26.6	25.8	25.4	25.0	24.9	24.6	24.9	26.0	26.7	26.7
	2% MCDB	28.9	29.4	29.4	28.8	27.6	27.0	26.8	26.5	26.5	26.8	27.9	28.4	28.4
	5% WB	26.3	26.5	26.6	26.1	25.4	24.8	24.5	24.2	24.1	24.4	25.5	26.2	26.2
	5% MCDB	28.4	29.0	28.8	28.2	27.2	26.4	26.2	25.9	26.0	26.4	27.5	28.0	28.0
	10% WB	25.9	26.1	26.2	25.7	25.0	24.3	24.0	23.6	23.6	23.8	25.0	25.7	25.7
	10% MCDB	28.0	28.6	28.3	27.8	26.8	26.0	25.7	25.4	25.6	26.0	27.1	27.5	27.5
(35) Mean Daily Temperature Range	MDBR	3.7	3.8	4.0	3.7	3.4	3.4	3.3	3.6	3.9	3.8	4.0	3.9	3.9
	5% DB	4.2	4.1	4.3	4.2	3.8	3.6	3.7	4.2	4.3	4.3	4.4	4.2	4.2
	MCWBR	2.4	2.2	2.4	2.4	2.3	2.2	2.4	2.5	2.3	2.4	2.3	2.2	2.2
	5% WB	3.8	3.9	4.2	3.9	3.5	3.3	3.5	3.9	4.2	4.1	4.0	3.9	3.9
	MCWBR	2.4	2.3	2.5	2.5	2.3	2.3	2.6	2.6	2.6	2.8	2.4	2.3	2.3
(40) Clear-Sky Solar Irradiance	taub	0.381	0.382	0.376	0.381	0.380	0.370	0.369	0.369	0.386	0.380	0.381	0.384	0.384
	taud	2.639	2.635	2.660	2.640	2.623	2.629	2.623	2.615	2.564	2.589	2.606	2.612	2.612
	Ebn at Noon	963	952	935	892	857	847	859	891	910	942	956	960	960
	Edh at Noon	101	100	94	90	85	82	84	91	102	103	103	103	103
(44) All-Sky Solar Radiation	RadAvg	6.18	6.07	5.67	4.86	4.13	3.94	4.12	4.86	5.67	6.09	6.40	6.29	6.29
	RadStd	0.53	0.45	0.46	0.35	0.36	0.22	0.21	0.28	0.31	0.52	0.47	0.53	0.53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page